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Period 7 APCS A

Group Name: Llama Games Inc. (working solo)

Project Title: Retro Archers

Description

- A 2D game with entities standing on platforms shooting arrows to defeat the enemy. Physics, health, and stamina are implemented. Entities can be defeated by knocking them off the platform or if they lose all their health from repeated shots or headshots. They can regenerate health and (for the player only) stamina by shooting apples with different buffs.
- **Functionalities:**
 - *Player System.* The player aims by clicking and dragging the mouse, with stamina draining the longer they hold. Releasing fires an arrow scaled to drag distance, spacebar triggers a jump (costing stamina), and keys 1–3 switch arrow types.
 - *Bot System.* Bots run a simple state machine that detects the player, aims with slight inaccuracy, shoots, and cools down. A wave spawner increases bot count after each cleared wave. Bots are not intelligent but will appear to learn over time. Each time they shoot an arrow, their wideness of their aim will start random but then be a little bit more constricted to the player's actual location over time until their aim width is basically the player's hitbox.
 - *Arrow System.* Three arrow types each behave differently, such as a standard arrow, a stun arrow, and a multi-shot arrow (subject to change). Arrows with special abilities cost more stamina. These arrows fly using velocity vectors with projectile motion applied, dealing damage and knockback on hit.
 - *Physics System.* Gravity is applied each frame to all entities, arrows, and apples, and collision checks handle landing on platforms. Arrow hits apply knockback that can push entities off the edges. Platforms aren't affected by any forces and will not move, only appear and disappear.
 - *Apples System.* Red, blue, and yellow apples spawn at the top of the screen and fall through it, giving the player limited time to shoot it as it falls. If they are shot they grant health, stamina, or both when shot by the player. Bot-fired arrows have no effect on apples.
 - *Stamina System.* Stamina regenerates slowly each frame and drains while aiming, jumping, or using certain arrow types. Shooting and jumping are blocked when it hits zero.
 - *Win/loss Conditions.* The player loses if their health reaches zero or they fall off-screen, and progresses by clearing waves of increasing size.
 - *Rendering and UI.* Sprites and shapes draw the player, enemies, arrows, apples, and platforms each frame. Displays health, stamina, current arrow type, and wave number, with start and game-over screens ending the gameplay.
- **Libraries needed:**

- PImage, PVector. Using PImage to load images of player, enemy, apple, arrow, etc. and of course PVector to implement physics and motion.
- **Anything else:**
 - Could use libraries for smooth animation or sound in the future. Could replace/add to PImage with the Sprite library. Not planning to add any libraries for now.

UML Diagram

Please see the last page.

How it works

Retro Archers is a 2D game where one player and bots standing on platforms try to knock each other off the platforms and into the abyss to win the game by shooting arrows at the enemy. If you shoot an enemy multiple times, they will lose all their health which is another way to win against them if they haven't been knocked off. There are multiple types of arrows with special abilities which will help the player win against the waves of enemies, who will usually be equipped with normal arrows.

These enemies, which are bots, will shoot arrows periodically and come in waves. Upon defeating an enemy, another will shortly spawn, and the more you defeat, the more that spawn in until you win the game. To prevent spamming arrows, players will have a stamina bar that regenerates slowly which will make it difficult to shoot arrows at a very fast rate in order to win easily. Enemies will start off shooting at you with little accuracy, but the longer they stay undefeated, the better their accuracy gets.

Apples will appear randomly which the players can shoot to gain health and/or stamina. A red apple will heal the player, a blue one will refill the player's stamina bar, and a yellow apple will regenerate the player's health and stamina. Nothing happens if a bot shoots an apple by accident.

Physics is implemented. The player will be able to use their stamina to jump up in the event that an arrow pushes them down. If the player loses all their health, they lose. If the player loses all their stamina, they must wait to regain some before shooting or jumping.

Keys for one player facing bots:

- Click and drag to aim (the longer you aim, the more stamina you lose at a linear rate). Release the mouse to shoot.
- Press the spacebar to jump. Unless you have already been knocked off, it will bring you to an upright, standing position and take away some stamina.
- You can press the 1, 2, and 3 keys to switch between different kinds of arrows, each with their own abilities.